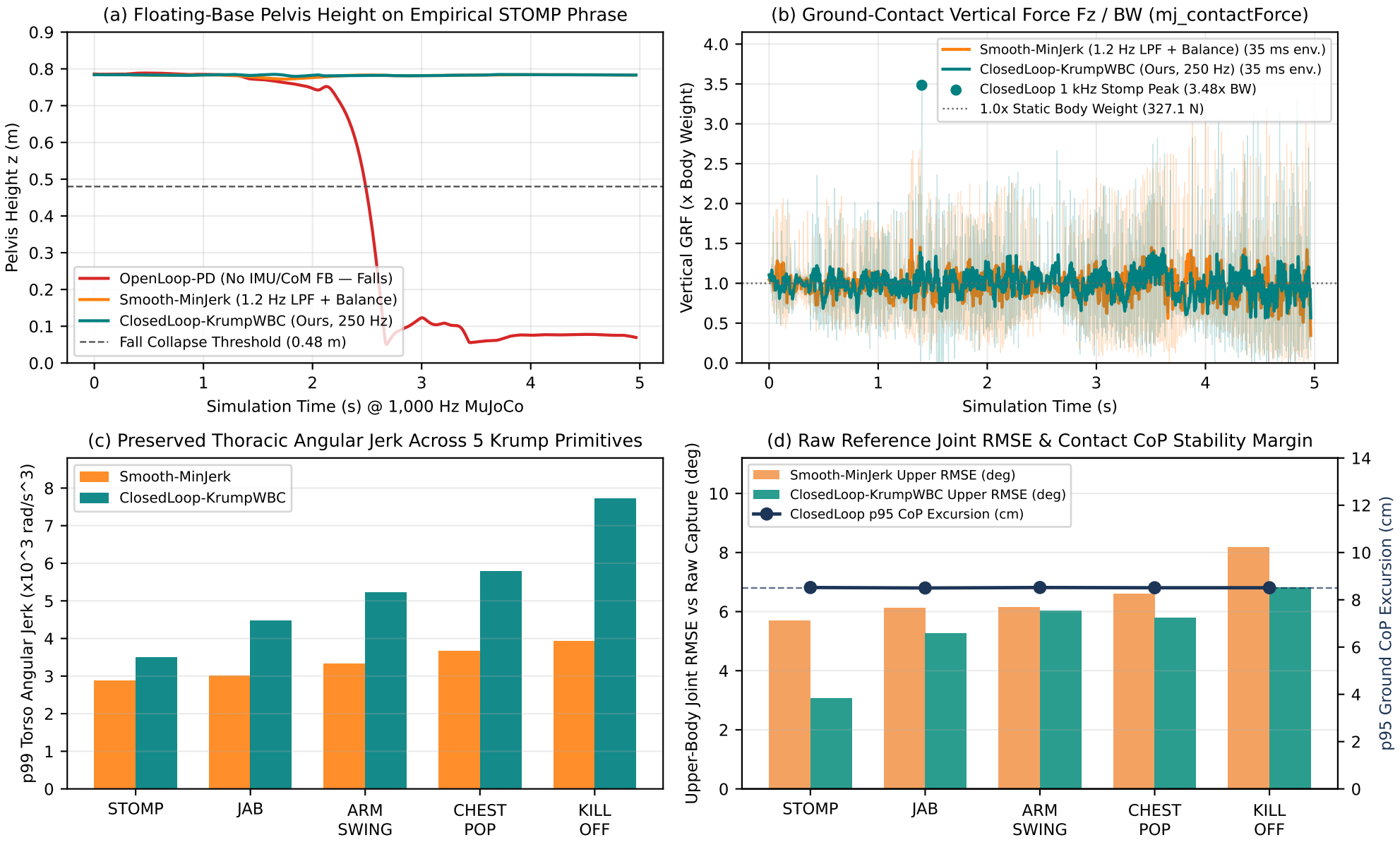

(a) Floating-Base Pelvis Height on Empirical STOMP Phrase
Pelvis Height z (m)
Simulation Time (s) @ 1,000 Hz MuJoCo
OpenLoop-PD (No IMU/CoM FB — Falls)
Smooth-MinJerk (1.2 Hz LPF + Balance)
ClosedLoop-KrumpWBC (Ours, 250 Hz)
Fall Collapse Threshold (0.48 m)
(b) Ground-Contact Vertical Force Fz / BW (mj_contactForce)
Vertical GRF (x Body Weight)
Simulation Time (s)
Smooth-MinJerk (1.2 Hz LPF + Balance) (35 ms env.)
ClosedLoop-KrumpWBC (Ours, 250 Hz) (35 ms env.)
ClosedLoop 1 kHz Stomp Peak (3.48x BW)
1.0x Static Body Weight (327.1 N)
(c) Preserved Thoracic Angular Jerk Across 5 Krump Primitives
p99 Torso Angular Jerk (x10^3 rad/s^3)
Smooth-MinJerk
ClosedLoop-KrumpWBC
STOMP
JAB
ARM SWING
CHEST POP
KILL OFF
(d) Raw Reference Joint RMSE & Contact CoP Stability Margin
Upper-Body Joint RMSE vs Raw Capture (deg)
p95 Ground CoP Excursion (cm)
Smooth-MinJerk Upper RMSE (deg)
ClosedLoop-KrumpWBC Upper RMSE (deg)
ClosedLoop p95 CoP Excursion (cm)
STOMP
JAB
ARM SWING
CHEST POP
KILL OFF